

Isobaric vapor–liquid equilibria for the binary systems 1-propyl alcohol + dipropyl ether and 1-butyl alcohol + dibutyl ether at 20 and 101.3 kPa

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Abstract

Isobaric vapor–liquid equilibrium measurements for the binary systems 1-propyl alcohol + dipropyl ether and 1-butyl alcohol + dibutyl ether are reported at 20 and 101.3 kPa. Both systems, which deviate positively from ideal behavior present a minimum boiling point azeotrope at both pressures, showing the azeotropic compositions a strong dependency on pressure. The activity coefficients and boiling points of the solutions were correlated with its composition by the Wilson, UNIQUAC, NRTL, and Wisniak–Tamir equations.

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1. Introduction

The distillation is a process very used in the separation and purification of organic compounds obtained in the chemical process industry. To be able to design the distillation columns appropriately is essential to have the best information on the vapor–liquid equilibrium (VLE) of the systems implied. This information involves isobaric equilibrium data as well as the presence of azeotropes and its variation with the pressure. Unfortunately, azeotropic data frequently are not available at different pressures. Therefore, determination of the VLE data of the azeotropic systems constitutes an interesting investigation work.

Aliphatic ethers, used extensively as solvents, are obtained normally by dehydration of the corresponding alcohol in the presence of an adequate catalyst. In many cases, the system formed by the ether and the homologous alcohol form an azeotropic mixture; therefore, final purification of aliphatic ethers in traditional technologies is a relative complex procedure. The separation can be improved by adding an agent that alters the relative volatility of the components (extractive distil-

lation) or making a simple change in pressure, provided that the azeotropic composition is sensitive to pressure (pressure swing distillation).

Some ethers like methyl 1,1-dimethylethyl ether (MTBE), ethyl 1,1-dimethylethyl ether (ETBE), methyl 1,1-dimethylpropyl ether (TAME), and diisopropyl ether (DIPE), used in the last years like oxygenated compound to reformulate gasoline in order to improve their octane rating, have been object of numerous studies and about them it has enough experimental information. Nevertheless, there are other ethers of which exists few experimental data. It deals with the ethers of high molecular weight and therefore of greater boiling point that have a great interest like solvent not excessively volatile. It is about compound as the dipropyl ether and dibutyl ether.

In this work it has been studied the binary systems formed for 1-propyl alcohol (PA) + dipropyl ether (DPE) and 1-butyl alcohol (BA) + dibutyl ether (DBE) at isobaric conditions. The present work was undertaken to measure the phase equilibrium properties (VLE data) of the two systems mentioned at 20 and 101.3 kPa, with the purpose of studying the influence of the pressure in the composition of the azeotropic mixture. This phenomenon can be very interesting in order to its separation by pressure swing distillation (PSD). It is well-known that, in some cases, changing the system pressure can affect the vapor–liquid

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equilibrium (VLE) of a mixture. This effect can be exploited to separate a binary mixture containing a minimum boiling azeotrope [1], provided that this mixture significantly changes composition over a moderate pressure range. Some VLE data are available in literature for these related mixtures [2], but at isothermal conditions.

The VLE data were found to be thermodynamically consistent at both pressures and were correlated with its composition by Wilson, UNIQUAC and NRTL equations.

2. Experimental

2.1. Chemicals

PA (>99.5 mass%, *Analytical grade*, Panreac), BA (>99.8 mass%, *Spectrophotometric grade*, Riedel-deHaën), DPE (90 mass %, *for synthesis*, Merck) and DBE (>99.5+ mass%, *puriss. p.a.* (GC). Fluka). DPE was purified to 99.5 mass% by batch distillation in a Fischer SPALTROHR-column HMS-500, controlled by a Fischer system D301-C. The other reagents were used without further purification after chromatography failed to show any significant impurities. The water content, determined using a Karl Fischer volumetric automatic titrator (Metrohm, 701 KF Titrino), was small in all chemicals (<0.05 mass%). Before measurements the liquids were dried over molecular sieves (Union Carbide, type 4 Å, 1/16 in. pellets). The densities and refractive indexes of pure liquids were measured at 298.15 K using an Anton Paar DMA 58 densimeter and an Abbe refractometer Atago 3T, respectively. Temperature was controlled to ± 0.01 K with a thermostated bath. The accuracy in density and refractive index measurements are $\pm 0.01 \text{ kg m}^{-3}$ and ± 0.0002 , respectively. The experimental values of these properties and the boiling points are given in Table 1 together with those given in the literature [3,4]. Appropriate precautions were taken when handling the reagents in order to avoid hydration.

2.2. Apparatus and procedure

The equilibrium vessel used in the measurements (Labodest VLE 602/D) was an all-glass dynamic-recirculating still equipped with a Cottrell circulation pump, manufactured by Fischer Labor und Verfahrenstechnik (Germany). The Cottrell pump ensures that both liquid and vapor phases are in intimate

contact during boiling and also in contact with the temperature sensing element. The equilibrium temperature was measured with a digital Hart Scientific thermometer model 1502A and a Pt 100 probe Hart Scientific model 5622 calibrated at the ENAC-accredited Spanish *Instituto Nacional de Técnica Aeroespacial*. The uncertainty is estimated to be ± 0.02 K. The temperature probe was calibrated against the ice and steam points of distilled water. A Fisher M101 phase equilibrium control system was used for measure and control the pressure and the heating power. The accuracy of measured pressures is estimated to be ± 0.1 kPa. The manometer was calibrated using the vapor pressure of ultrapure water.

In each experiment, the pressure was fixed and the heating and shaking system of the liquid mixture was turned on. The still was operated under constant pressure until equilibrium was reached. Equilibrium conditions were assumed when constant temperature and pressure were obtained for 30 min or longer. Then, samples of liquid and condensate vapor were taken for analysis. The sample extractions were carried out with special syringes that allowed withdrawal of small volume samples.

2.3. Analysis

Composition of the sampled liquid and condensed vapor phase were determined using a CE Instruments GC 8000 Top gas chromatograph (GC), after calibration with gravimetrically prepared standard solutions. A flame ionization detector was used together with a 30 m, 0.454 mm i.d., capillary column, (DB-MTBE, J & Scientific). The GC response peaks were treated with Chrom-Card for Windows. Column, injector and detector temperatures were 363, 473, and 498 K respectively, for all systems. Very good peak separation was achieved under these conditions and calibration analyses were carried out to convert the peak area ratio to the mass composition of the sample. At least two analyses were made of each liquid and vapor composition. The standard deviation in the mole fraction was usually less than 0.001.

3. Results and discussion

The pure component vapor pressure P_i^0 for PA and BA were taken from Refs. [5,6]. For DPE and DBE, pure component vapor pressures were determined experimentally as a function of the temperature, using the same equipment as that for obtaining the

Table 1
Densities d , refractive index n_D , and normal boiling points T_b of pure components

Component	d (kg m^{-3}) (298.15 K)		n_D (298.15 K)		T_b (K)	
	Experimental	Literature ^a	Experimental	Literature ^a	Experimental	Literature ^b
Propyl alcohol (PA)	799.62	799.51	1.3836	1.3837	369.75	370.35
Dipropyl ether (DPE)	741.81	742.00	1.3784	1.3780	363.16	362.79
Butyl alcohol (BA)	805.65	805.77	1.3976	1.3971	390.75	390.81
Dibutyl ether (DBE)	764.04	764.10	1.3970	1.3968	413.67	413.44

^a Taken from TRC tables [3].

^b Taken from Daubert and Danner [4].

Table 2

Experimental vapor pressures (P_i^0) for dipropyl ether (DPE) and dibutyl ether (DBE)

DPE		DBE	
T (K)	P (kPa)	T (K)	P (kPa)
302.60	10.47	346.03	10.04
306.36	12.43	350.43	12.03
310.41	14.95	354.18	14.02
313.39	16.91	359.28	17.10
315.97	18.97	363.41	20.02
318.45	20.97	367.17	23.04
320.69	22.93	370.72	26.18
323.79	25.95	373.59	28.95
326.58	28.97	376.37	31.96
329.15	31.95	379.91	35.96
331.52	34.95	383.09	39.94
333.76	37.95	386.03	43.95
335.79	40.90	388.78	47.95
337.80	43.95	391.29	51.85
339.62	46.85	393.59	55.85
341.27	49.70	395.84	59.90
343.07	52.91	398.48	64.90
345.22	56.95	400.94	69.92
347.19	60.85	403.18	74.81
349.07	64.87	405.40	79.93
351.36	69.90	407.43	84.78
353.45	74.84	409.20	89.87
355.48	79.91	411.12	94.88
357.37	84.80	413.11	100.12
359.21	89.75	413.67	101.30
360.93	94.65		
363.16	101.30		

VLE data. The pertinent results appear in Table 2. The measured vapor pressures were correlated using the Antoine equation

$$\ln P_i^0 \text{ (kPa)} = A_i - \frac{B_i}{T \text{ (K)} - C_i} \quad (1)$$

Table 3

Vapor pressure parameters

Compound	Eq ^a .	A_i	B_i	C_i	D_i	E_i	Reference
Propyl alcohol (PA)	$\ln P^0 \text{ (kPa)} = A - B/[T \text{ (K)} - C]$	16.0353	3415.56	70.733			b
Dipropyl ether (DPE)	$\ln P^0 \text{ (kPa)} = A - B/[T \text{ (K)} - C]$	14.1782	3018.27	47.440			This work ^c
	$\log P^0 \text{ (kPa)} = A - B/[T \text{ (K)} - C]$	6.0643	1245.19	56.138			d
	$\ln P^0 \text{ (Pa)} = A + B/T \text{ (K)} + C \ln T + DT^E$	106.26	-7557.0	-12.792	1.123×10^{-5}	2	e
	$\log P^0 \text{ (mmHg)} = A - B/[T \text{ (°C)} + C]$	6.9476	1256.50	219.013			f
Butyl alcohol (BA)	$\ln P^0 \text{ (kPa)} = A - B/[T \text{ (K)} - C]$	15.8219	3499.92	78.111			g
Dibutyl ether (DBE)	$\ln P^0 \text{ (kPa)} = A - B/[T \text{ (K)} - C]$	14.7309	3681.13	49.728			This work ^c
	$\ln P^0 \text{ (kPa)} = A - B/[T \text{ (K)} - C]$	13.8340	3137.49	74.48			h
	$\ln P^0 \text{ (Pa)} = A + B/T \text{ (K)} + C \ln T + DT^E$	74.112	-7694.6	-7.306	8.795×10^{-18}	6	e
	$\log P^0 \text{ (mmHg)} = A - B/[T \text{ (°C)} + C]$	6.7963	1297.29	191.03			f

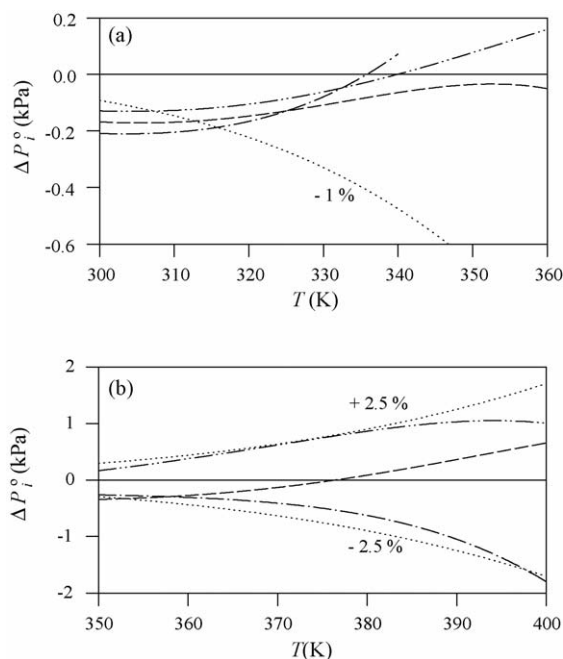
^a Vapor pressure equations.^b Taken from Aucejo et al. [5].^c Antoine's parameters were calculated from the experimental data in Table 2.^d Parameters taken from Antosik et al. [7].^e Parameters taken from Daubert and Danner [4].^f Parameters taken from Boublik et al. [8].^g Parameters taken from Dejoz et al. [6].^h Parameters taken from Paul et al. [9].

Fig. 1. Differences, $\Delta P_i^0 = P_{i,\text{exptl}}^0 - P_{i,\text{lit}}^0$, obtained for vapour pressures using the parameters given in Table 3. (a) Dipropyl ether: Antosik et al. [7] (---); Daubert and Danner [4] (---); Boublik et al. [8] (---). (b) Dibutyl ether: Paul et al [9] (---); Daubert and Danner [4] (---); Boublik et al. [8] (---).

whose parameters A_i , B_i , and C_i are reported in Table 3 together with some literature values [4,7,8,9]. The deviations, $\Delta P_i^0 = P_{i,\text{exptl}}^0 - P_{i,\text{lit}}^0$, calculated by means of the Antoine equation using the constant values from Table 3 have been graphically represented in Fig. 1.

The temperature T , the liquid-phase x_i and vapor-phase y_i mole fractions at 20 and 101.3 kPa are reported in Tables 4–7, and in Figs. 2 and 3. The activity coefficients γ_i for the systems PA + DPE and BA + DBE, were calculated from the following

Table 4

Experimental vapor–liquid equilibrium data for propyl alcohol (1) + dipropyl ether (2) at 20 kPa

<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>γ</i> ₁	<i>γ</i> ₂
317.33	0.000	0.000		1.000
316.35	0.058	0.091	3.704	1.006
315.86	0.152	0.165	2.655	1.048
315.84	0.210	0.193	2.250	1.088
315.91	0.268	0.216	1.965	1.137
316.09	0.327	0.236	1.742	1.197
316.41	0.406	0.260	1.522	1.294
316.79	0.471	0.281	1.385	1.391
317.20	0.524	0.308	1.334	1.462
317.77	0.583	0.326	1.228	1.589
318.31	0.629	0.348	1.179	1.689
318.94	0.674	0.375	1.144	1.795
319.71	0.720	0.407	1.115	1.922
320.70	0.764	0.444	1.085	2.051
321.84	0.806	0.487	1.061	2.198
323.33	0.848	0.547	1.045	2.333
324.89	0.886	0.611	1.029	2.516
326.82	0.923	0.694	1.013	2.732
329.05	0.957	0.802	1.006	2.922
331.22	0.984	0.919	1.004	2.859
332.63	1.000	1.000	1.000	

Table 6

Experimental vapor–liquid equilibrium data for butyl alcohol (1) + dibutyl ether (2) at 20 kPa

<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>γ</i> ₁	<i>γ</i> ₂
363.51	0.000	0.000		1.000
362.77	0.011	0.052	2.881	0.982
360.85	0.030	0.129	2.717	0.989
359.17	0.052	0.196	2.586	0.996
357.21	0.083	0.276	2.495	1.000
355.56	0.119	0.342	2.321	1.010
354.24	0.157	0.396	2.171	1.020
353.11	0.197	0.438	2.017	1.042
352.13	0.256	0.481	1.779	1.081
351.06	0.336	0.538	1.594	1.126
350.40	0.393	0.556	1.457	1.214
350.06	0.451	0.585	1.355	1.274
349.75	0.509	0.611	1.272	1.352
349.48	0.569	0.637	1.202	1.452
349.38	0.631	0.667	1.141	1.562
349.32	0.690	0.697	1.094	1.695
349.36	0.750	0.734	1.056	1.847
349.49	0.809	0.776	1.030	2.018
349.85	0.870	0.830	1.006	2.225
350.30	0.934	0.899	0.994	2.547
351.18	1.000	1.000	1.000	

equation [10]:

$$\gamma_i = \frac{y_i P}{x_i P_i^0} \quad (2)$$

where *P* is the total pressure. In Eq. (2), the vapor phase is assumed to be an ideal gas and the pressure dependence of the liquid phase fugacity is neglected. Eq. (2) was selected to calculate activity coefficients because the low pressures observed in the present VLE data makes this simplification reasonable.

The results reported in Tables 4–7 indicate that the measured systems exhibit positive deviations from ideal behavior.

From the calculated activity coefficients, the excess Gibbs energy could then be evaluated using the equation

$$G^E = RT(x_1 \ln \gamma_1 + x_2 \ln \gamma_2) \quad (3)$$

Figs. 4 and 5 show the excess Gibbs energy for the liquid phase at 20 and 101.3 kPa for both systems, from which it is concluded that only positive deviations from ideality are present

Table 5

Experimental vapor–liquid equilibrium data for propyl alcohol (1) + dipropyl ether (2) at 101.3 kPa

<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>γ</i> ₁	<i>γ</i> ₂
363.16	0.000	0.000		1.000
361.38	0.051	0.096	2.617	1.006
360.07	0.104	0.168	2.386	1.020
359.17	0.166	0.229	2.108	1.045
358.65	0.234	0.280	1.872	1.079
358.45	0.296	0.316	1.683	1.123
358.38	0.354	0.344	1.537	1.176
358.44	0.413	0.376	1.431	1.231
358.66	0.474	0.401	1.322	1.307
358.94	0.529	0.429	1.253	1.379
359.29	0.582	0.456	1.193	1.465
359.84	0.636	0.489	1.143	1.555
360.44	0.688	0.521	1.098	1.671
361.25	0.742	0.561	1.063	1.802
362.09	0.787	0.602	1.038	1.933
363.25	0.836	0.654	1.014	2.108
364.69	0.884	0.717	0.993	2.328
365.64	0.910	0.764	0.990	2.427
366.91	0.940	0.829	0.990	2.556
368.50	0.975	0.911	0.986	3.050
369.75	1.000	1.000	1.000	

Table 7

Experimental vapor–liquid equilibrium data for butyl alcohol (1) + dibutyl ether (2) at 101.3 kPa

<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>γ</i> ₁	<i>γ</i> ₂
413.67	0.000	0.000		1.000
411.08	0.025	0.101	2.054	0.993
408.39	0.050	0.181	1.988	1.002
405.77	0.082	0.265	1.929	1.003
403.47	0.116	0.334	1.837	1.011
401.90	0.141	0.385	1.837	1.006
398.58	0.196	0.475	1.828	1.013
396.73	0.250	0.523	1.679	1.044
395.51	0.289	0.557	1.616	1.062
394.47	0.339	0.590	1.512	1.092
393.78	0.389	0.622	1.423	1.113
393.09	0.436	0.645	1.350	1.157
392.49	0.492	0.670	1.267	1.218
391.74	0.555	0.700	1.207	1.292
391.21	0.621	0.729	1.145	1.392
390.98	0.677	0.759	1.102	1.465
390.51	0.739	0.788	1.065	1.622
390.33	0.800	0.824	1.035	1.773
390.28	0.863	0.868	1.014	1.927
390.41	0.929	0.919	0.992	2.292
390.75	1.000	1.000	1.000	

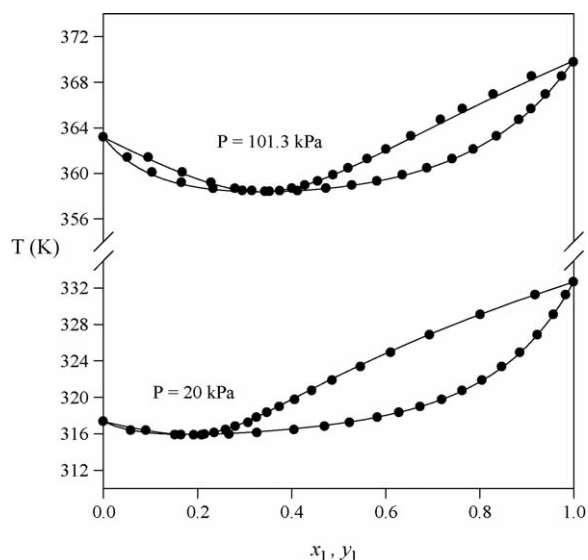


Fig. 2. Experimental data for the system 1-propyl alcohol (1) + dipropyl ether (2) at 20 and 101.3 kPa. Experimental data (●). Smoothed line calculated using the Wilson model with the parameters given in Table 10.

in the liquid phase. Moreover, values for the excess Gibbs energy of PA + DPE and BA + DBE systems decrease as the pressure increases for a given mole fraction. Maximum G^E values range from $856.26 \text{ J mol}^{-1}$ at 20 kPa to $815.46 \text{ J mol}^{-1}$ at 101.3 kPa for the PA + DPE system and from $784.66 \text{ J mol}^{-1}$ at 20 kPa to $728.05 \text{ J mol}^{-1}$ at 101.3 kPa for the BA + DBE system. These maxima appear at approximately the same mole fraction ($x_1 \approx 0.5$, regular solution) for all of the pressures and systems studied.

Minimum boiling azeotropes are present in both systems at 20 and 101.3 kPa. The compositions and boiling temperatures of the azeotropes are summarised in Table 8, together with the corresponding literature values. In all cases, the azeotropic com-

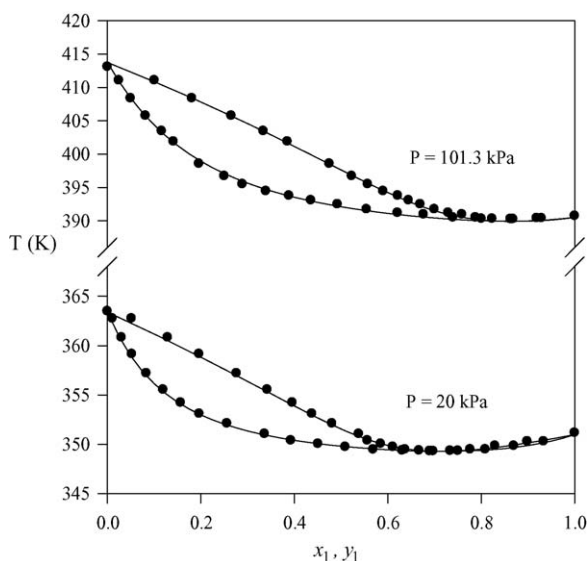


Fig. 3. Experimental data for the system 1-butyl alcohol (1) + dibutyl ether (2) at 20 and 101.3 kPa. Experimental data (●). Smoothed line calculated using the Wilson model with the parameters given in Table 10.

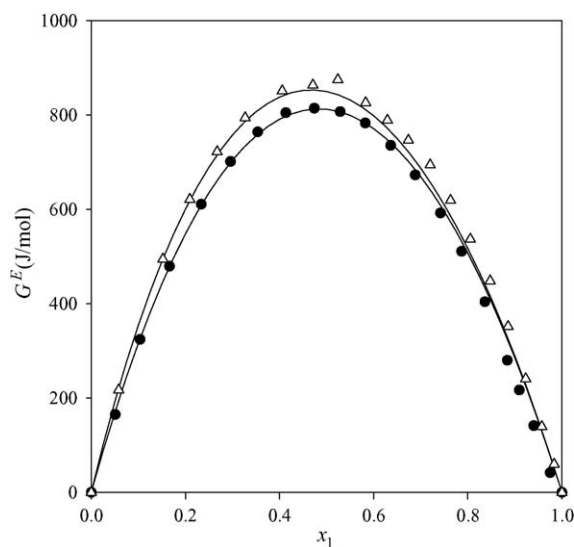


Fig. 4. Excess Gibbs energy function for the 1-propyl alcohol (1) + dipropyl ether (2) system. $P = 101.3 \text{ kPa}$ (●); $P = 20 \text{ kPa}$ (Δ). Smoothed line calculated using the Wilson model with the parameters given in Table 10.

position was obtained from the best polynomial fit of the function $(y_1 - x_1) = f(x_1)$. Azeotropic temperatures were obtained from the best polynomial fit for $T = f(x_1)$, using the x_1 values previously determined. In Table 8, it can be observed the strong dependency of azeotropic compositions on pressure.

The VLE data reported in this work were found to be thermodynamically consistent by the point-to-point method of Van Ness [12], as modified by Fredenslund [13]. For both systems, at both pressures, consistency was achieved using a three parameters Legendre polynomial. These parameters, together with the pertinent statistics required by the Fredenslund test, are shown in Table 9 for these systems at both pressures.

The activity coefficients were correlated with the Wilson, NRTL, and UNIQUAC equations. The parameters of these equa-

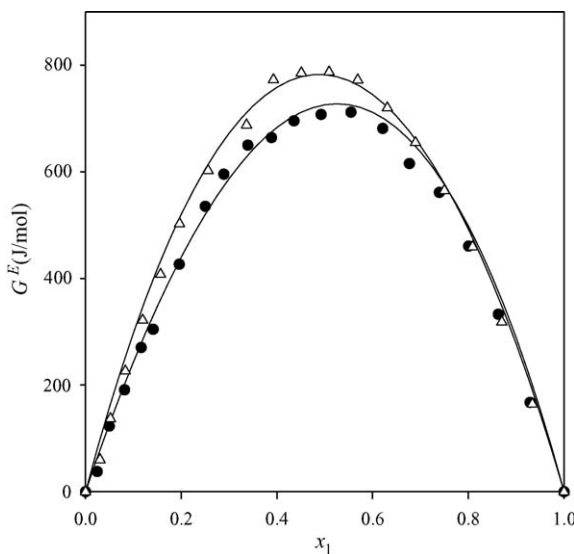


Fig. 5. Excess Gibbs energy function for the 1-butyl alcohol (1) + dibutyl ether (2) system. $P = 101.3 \text{ kPa}$ (●); $P = 20 \text{ kPa}$ (Δ). Smoothed line calculated using the Wilson model with the parameters given in Table 10.

Table 8
Azeotropic compositions and temperatures

System	<i>P</i> (kPa)	<i>T</i> (K)		<i>x</i> ₁	
		Experimental	Literature ^a	Experimental	Literature ^a
PA (1) + DPE (2)	20	315.75	–	0.1771	–
	101.3	358.31	358.85	0.3353	0.4098
BA (1) + DBE (2)	20	349.45	–	0.7087	–
	101.3	390.29	390.59	0.8858	0.8754

^a Taken from Gmehling et al. [11].

Table 9
Consistency test for the experimental systems considered in VLE measurements

System	<i>P</i> (kPa)	<i>L</i> ₁ ^a	<i>L</i> ₂ ^a	<i>L</i> ₃ ^a	AADy ₁ ^b	AADP ^c (kPa)
PA (1) + DPE (2)	20.0	1.3017	−0.2132	0.0203	0.0067	0.06
	101.3	1.0790	−0.1110	−0.0079	0.0061	0.18
BA (1) + DBE (2)	20.0	1.0482	−0.0341	−0.0517	0.0035	0.06
	101.3	0.8410	0.0583	−0.1059	0.0044	0.39

^a Legendre parameters.

^b Average absolute deviation in vapor phase composition.

^c Average absolute deviation in pressure.

Table 10
Parameters and deviations between experimental and calculated values for different *G*^E models

Model	A_{12}^0 (J mol ^{−1})	A_{12}^1 (J mol ^{−1} K ^{−1})	A_{21}^0 (J mol ^{−1})	A_{21}^1 (J mol ^{−1} K ^{−1})	α	AADy ₁ ^a	AADT ^b (K)
PA + DPE system							
Wilson ^c	4515.54	−0.09	2982.80	−9.78	0.3	0.0045	0.06
UNIQUAC ^d	107.71	−2.69	2122.55	1.09		0.0055	0.07
NRTL	2209.81	−3.48	3426.70	−2.18		0.0051	0.07
BA + DBE system							
Wilson ^c	12498.93	−23.73	−5792.23	15.39	0.3	0.0029	0.20
UNIQUAC ^d	−29.74	−2.24	2138.09	−0.03		0.0051	0.20
NRTL	−3783.03	14.48	4715.44	−8.69		0.0040	0.20

^a Average absolute deviation in temperature.

^b Average absolute deviation in vapor phase composition.

^c Liquid volumes from DECHEMA [14].

^d Volume and surface parameters from DECHEMA [14].

tions were obtained by minimizing the following objective function (OF)

$$\text{OF} = \sum_{i=1}^N \left(\left| \frac{T_i^{\text{exptl}} - T_i^{\text{calc}}}{T_i^{\text{exptl}}} \right| + |y_i^{\text{exptl}} - y_i^{\text{calc}}| \right) \quad (4)$$

and are reported in Table 10, together with the pertinent statistics of VLE correlation. Simultaneous correlation of experimental

data at 20 and 101.3 kPa was carried out for each system, and for that reason, to take into account the strong dependence of the excess Gibbs energy on temperature, the parameters for all these equations (*A*_{*ij*}, *A*_{*ji*}) were considered to be linear function of the temperature [15,16]

$$A_{ij} = A_{ij}^0 + A_{ij}^1 T \quad (5)$$

Table 11
Coefficients in correlation of boiling points, Eq. (6). average deviation, root mean square deviation and maximum deviation in temperature

System	<i>P</i> (kPa)	<i>C</i> ₀	<i>C</i> ₁	<i>C</i> ₂	AADT ^a (K)	RMSD ^b (K)	Max. AADT ^c
PA (1) + DPE (2)	20	−31.46	−17.23	−22.24	0.09	0.046	0.25
	101.3	−30.61	−2.19	−16.38	0.04	0.022	0.13
BA (1) + DBE (2)	20	−29.78	19.90	−24.76	0.19	0.095	0.43
	101.3	−38.78	26.00	−19.66	0.13	0.071	0.46

^a Average absolute deviation in temperature.

^b Root mean square deviation.

^c Maximum deviation.

The results given in Table 10 show that all models fitted very well the experimental data of these systems.

The boiling point temperatures of each system at 20 and 101.3 kPa were well correlated with mole fractions by the equation proposed by Wisniak and Tamir [17]

$$T(\text{K}) = x_i T_i^0(\text{K}) + x_j T_j^0(\text{K}) + x_i x_j \sum_{k=0}^m C_k (x_i - x_j)^k \quad (6)$$

In this equation $T_i^0(\text{K})$ is the boiling point of the pure component i , and m is the number of terms used in the series expansion of $(x_i - x_j)$. The various constants of Eq. (6) are reported in Table 11, with information indicating the goodness of fit of the correlation.

List of symbols

A_i	Antoine's equation parameter, Eq. (2)
B_i	Antoine's equation parameter, Eq. (2)
C_i	Antoine's equation parameter, Eq. (2)
G^E	excess Gibbs energy (J mol^{-1})
L_k	parameters of Legendre polynomial
P	total vapor pressure of mixture (kPa)
P^0	pure component vapor pressure (kPa)
R	universal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)
T	absolute temperature (K)
x, y	compositions of the liquid and vapor phases

Greek letter

γ	activity coefficient
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Subscript

i, j	components i and j
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